RMIT Classification: Trusted

# COSC2626/2640
# Introduction to Cloud Computing

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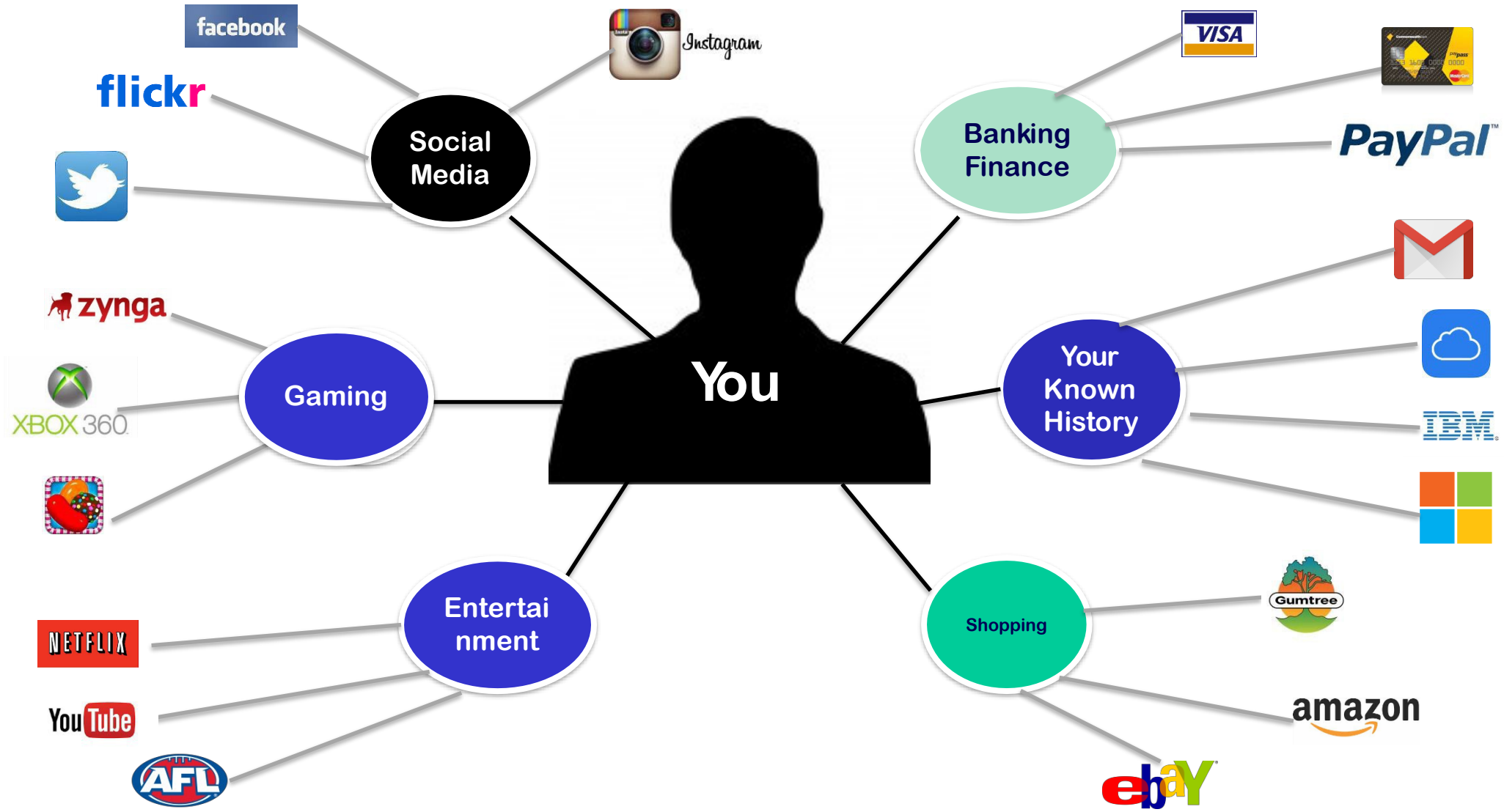
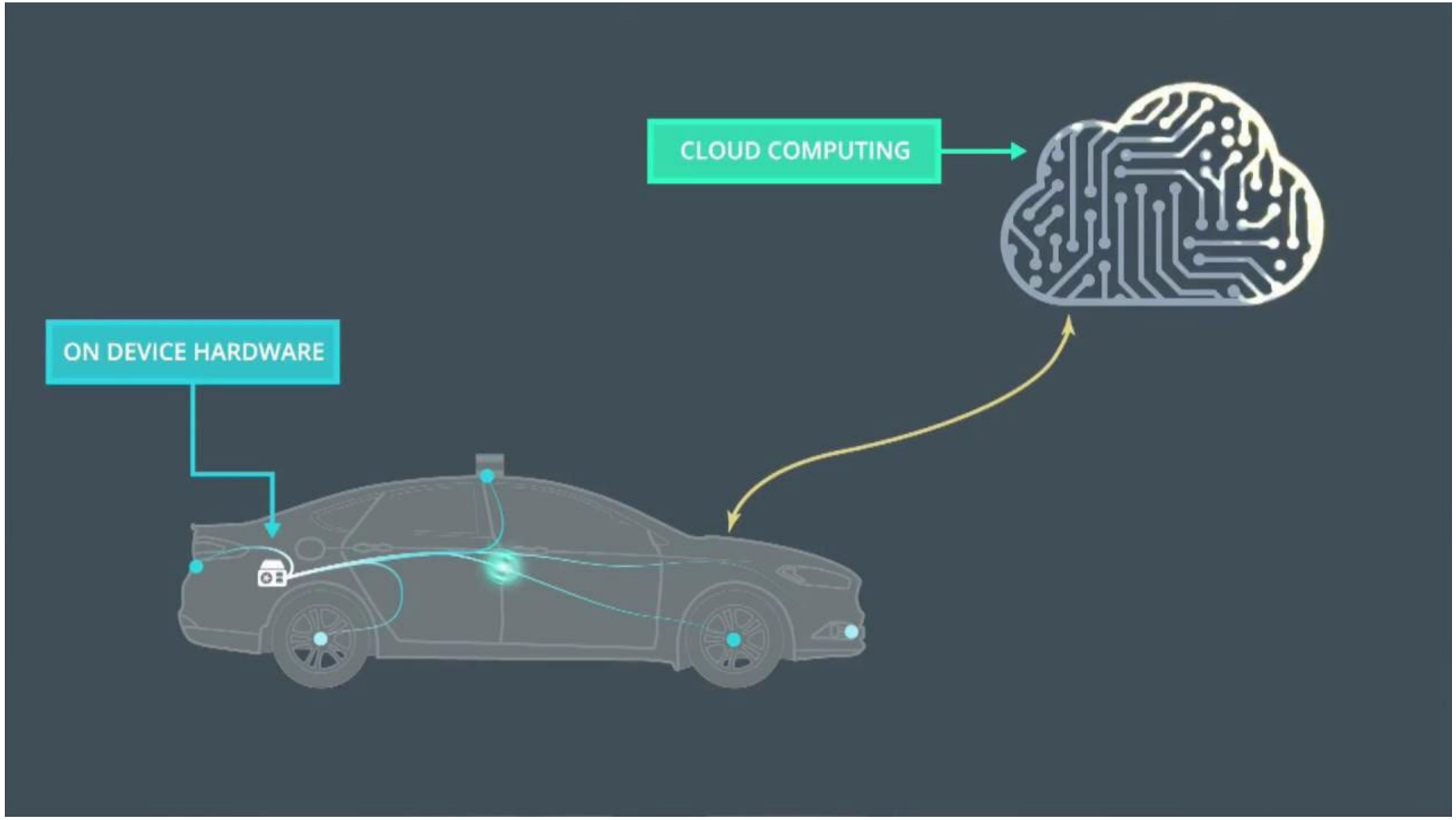
# This Lecture

- Cloud computing use cases
- Cloud computing and relevant computings
- Cloud characteristics
- Cloud service models
- Cloud deployment models

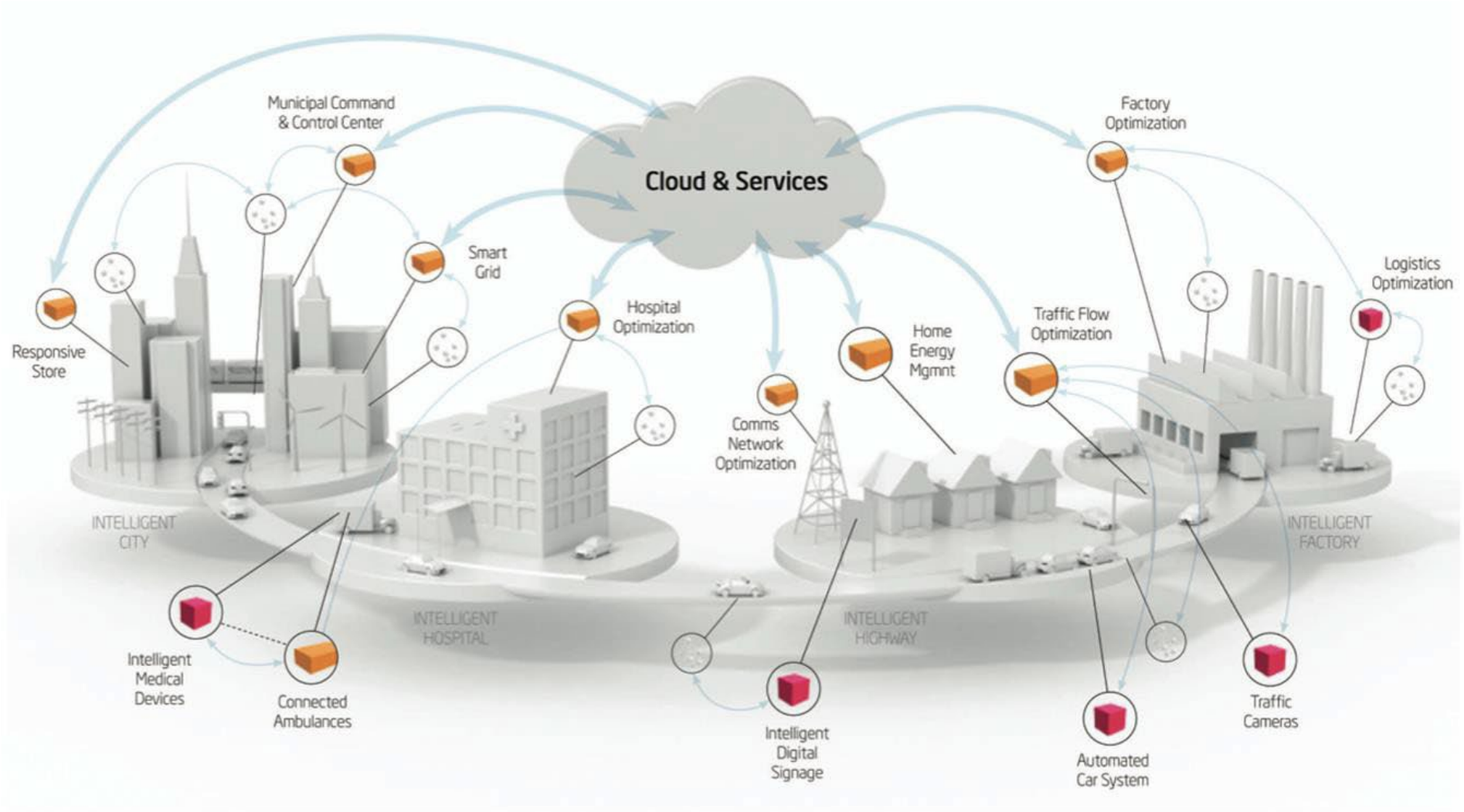
# Cloud computing use cases



# Use Case – Cloud accelerates AI: Self-driving car



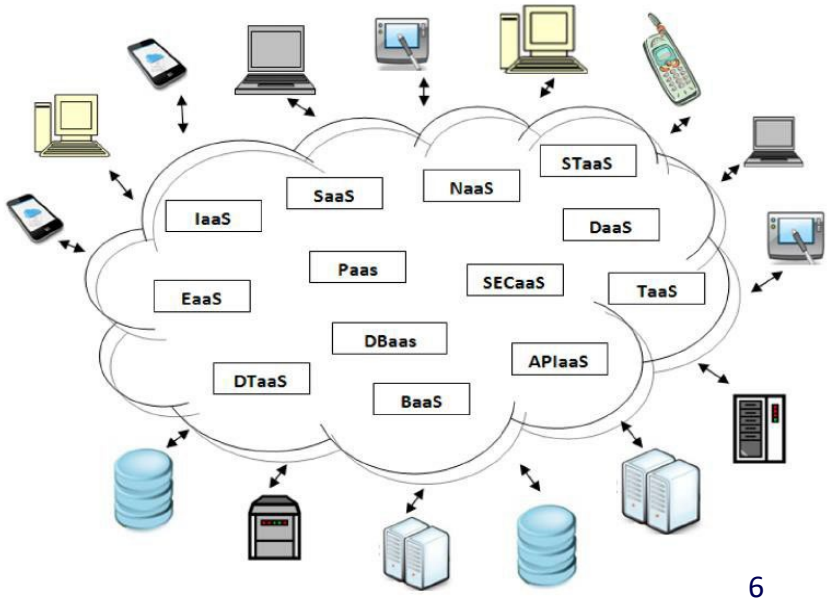
# Use Case – Cloud accelerates AI: Smart city



# Cloud Computing

- The delivery of computing services over the Internet

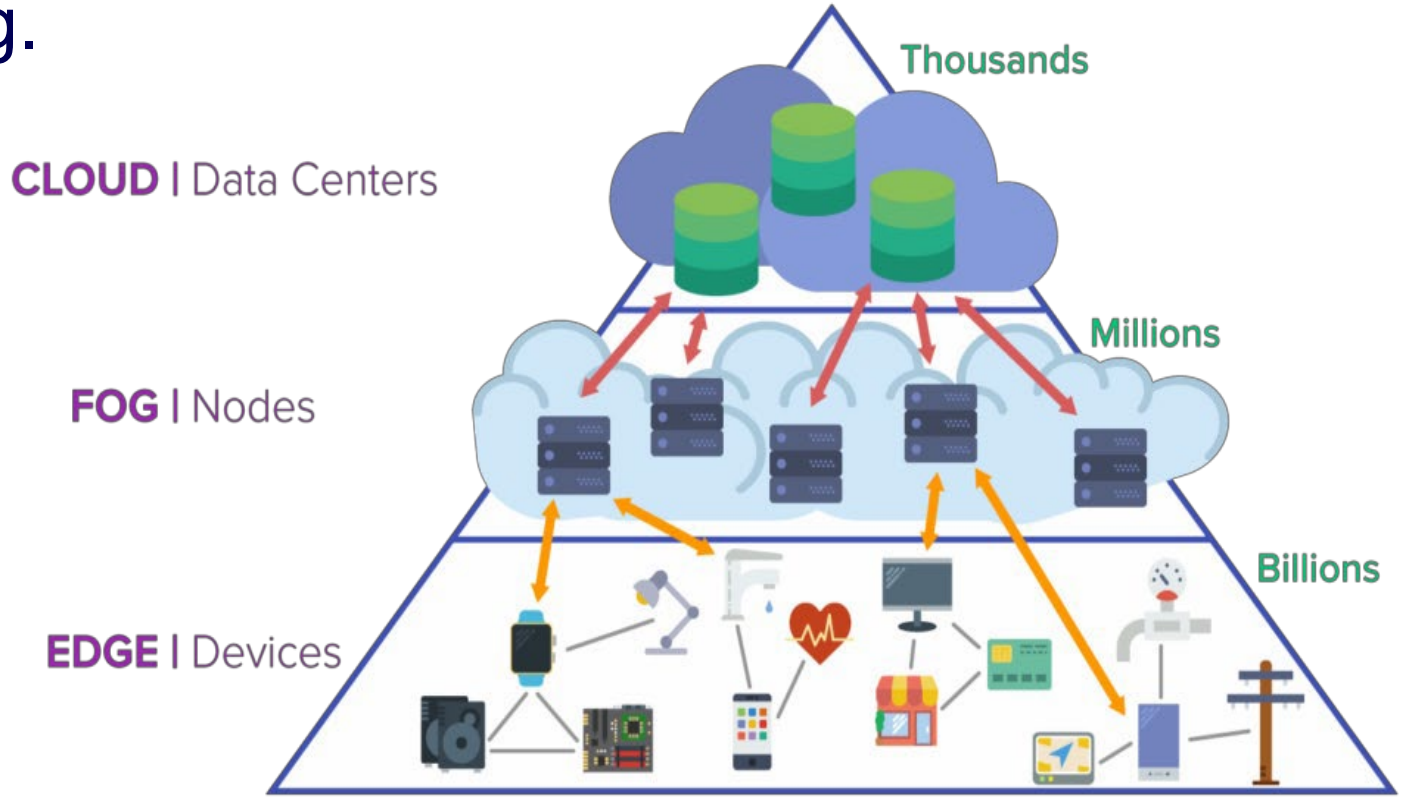
- Cloud is built over data centres that are centralized or distributed.
- Applies parallel or distributed computing or combination of both
- On-demand and elastic provisioning through virtualization




- What is your view of Cloud Computing?

# Fog computing

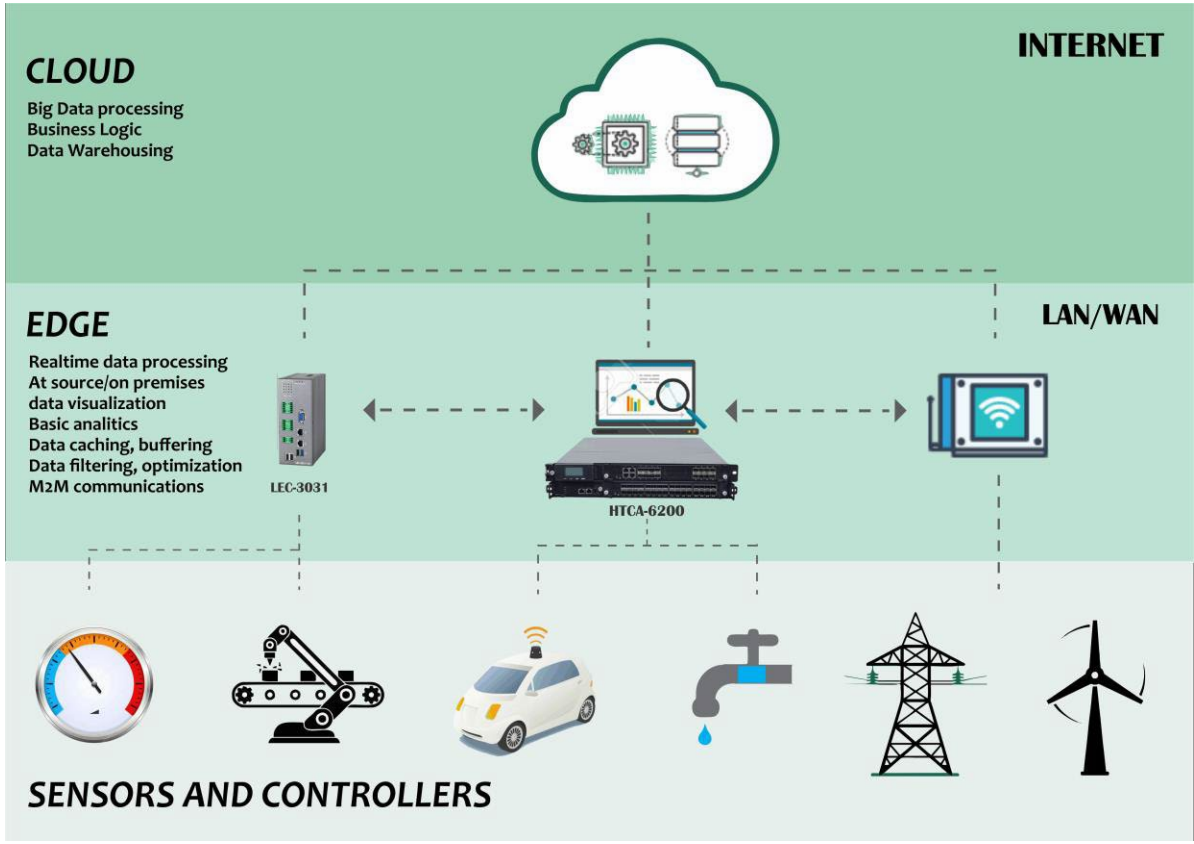
- A small cloud before data go the actual cloud: enterprise network
- Provide computing, storage and networking services for data processing.



# Edge computing

- Local processing with/without the cloud involved
- Data from IoT devices analysed at the edge: devices themselves, base stations or gateways

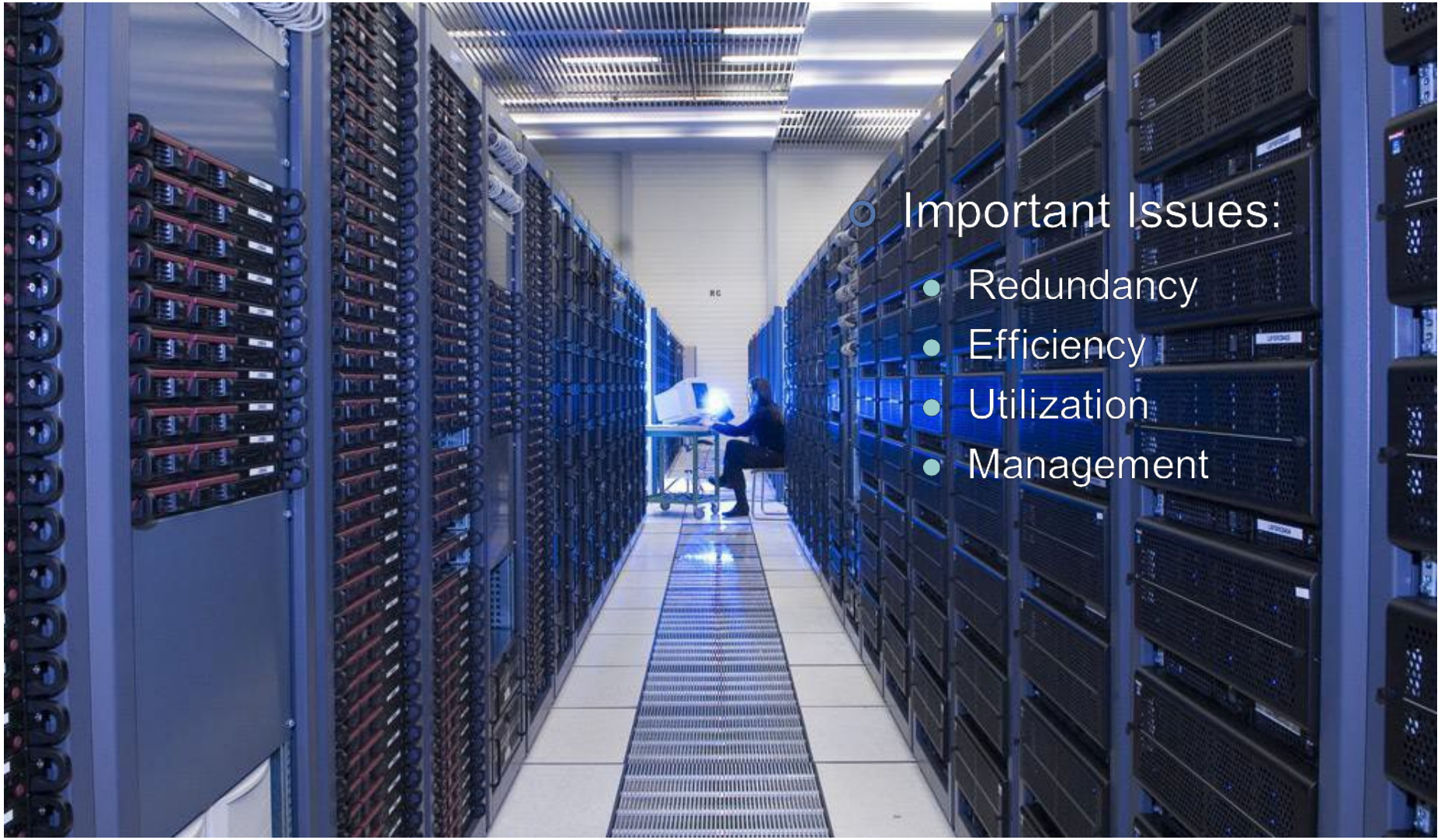
- Why do we want to do this?



# Essential cloud characteristics

- Large data centres
- Virtualization
- Resource pooling
- Elastic provisioning
- Self-service and on-demand services
- Pay-as-you go
- Measured service
- ….

# Large Data Centres

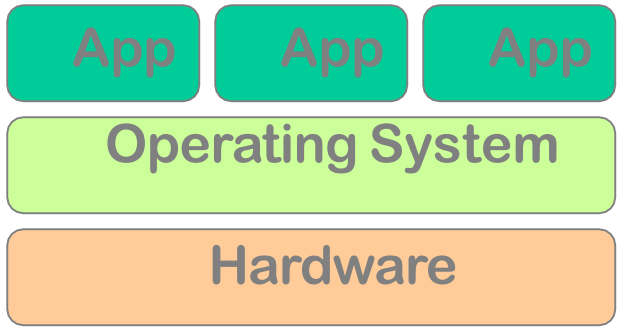


- Important Issues:
  - Redundancy
  - Efficiency
  - Utilization
  - Management

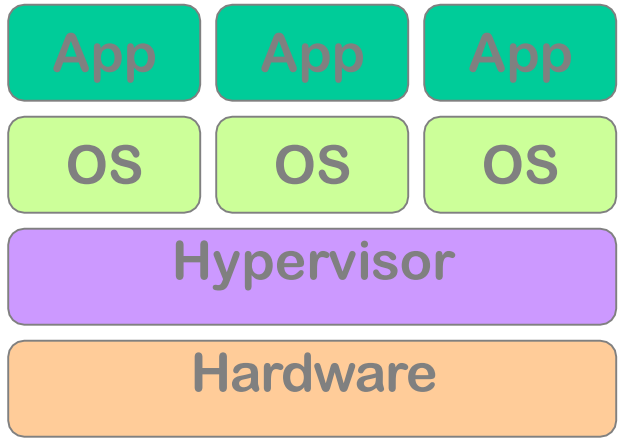
# Large Data Centres

- Data centres have grown rapidly in recent years.
- Amazon, Microsoft, Google, Alibaba, etc. are all companies have invested billions of dollars in datacentre construction and automation.
- Huge volumes of hardware, software, and database resources in these data centres can be allocated dynamically to millions of Internet users simultaneously.
- Guaranteed QoS and cost-effectiveness.

# Virtualization




Traditional computing environment

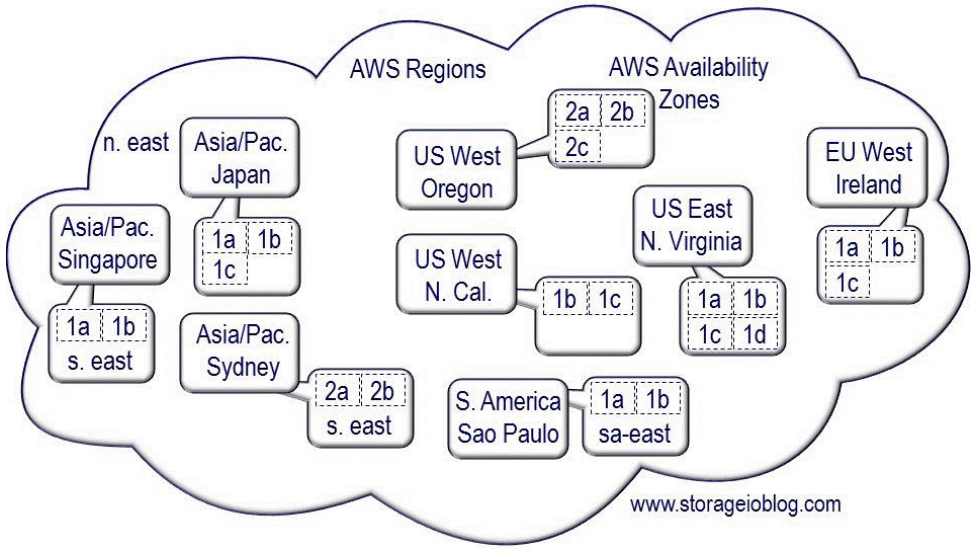


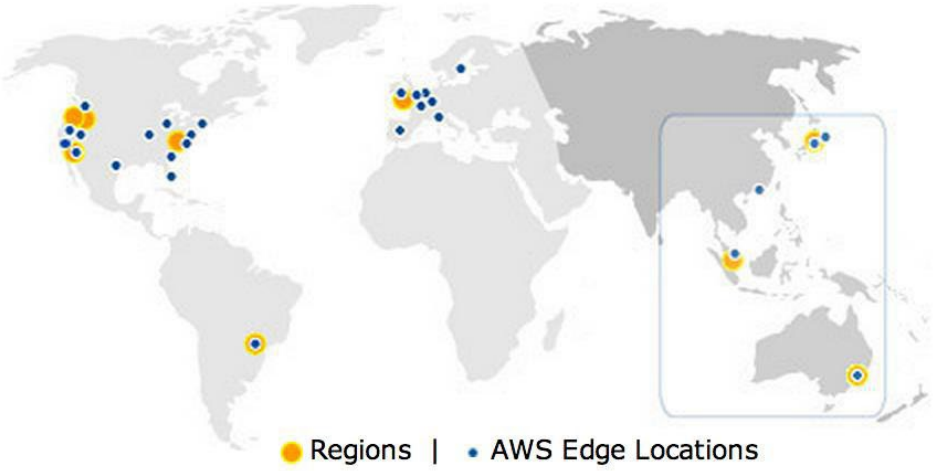

Virtualized computing environment

A hypervisor or virtual machine monitor (VMM) is a piece of computer software, firmware or hardware that creates and runs virtual machines.

# Resource Pooling

- The provider's computing resources are pooled to serve multiple consumers using a multi-tenant model, with different physical and virtual resources dynamically assigned and reassigned according to consumer demand
- Examples: AWS availability zones

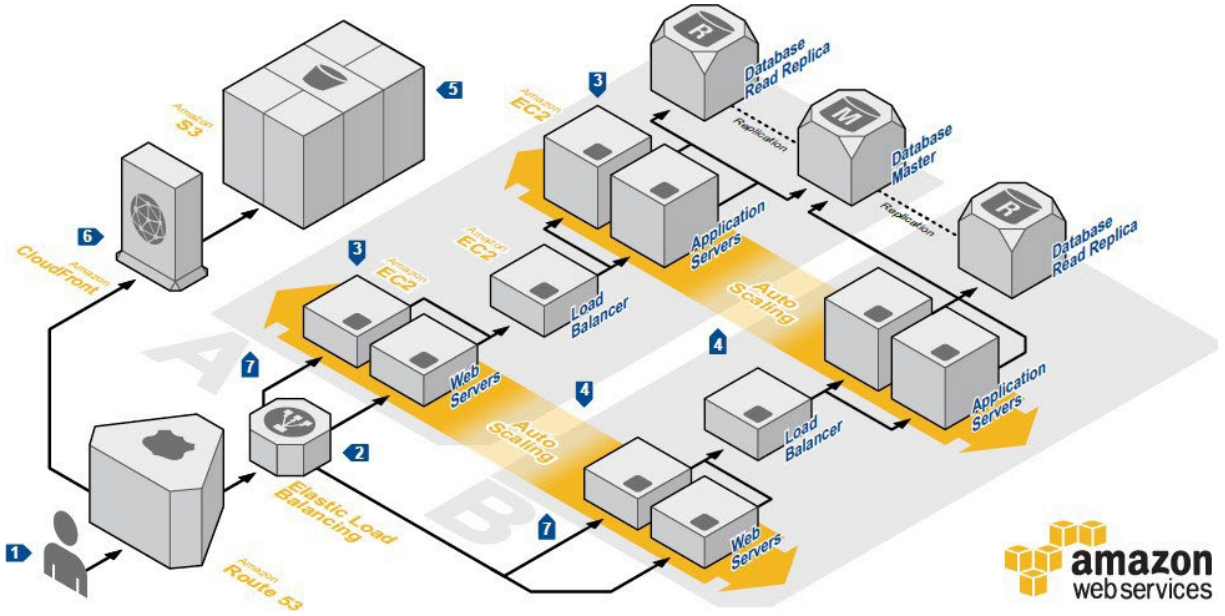


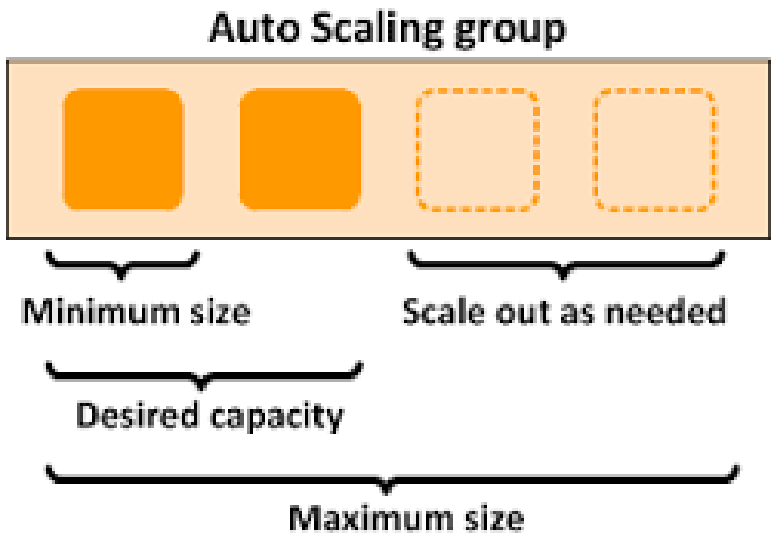



# Elastic Provisioning

- Capabilities can be elastically provisioned and released, to scale rapidly outward and inward with demand

- Examples: AWS auto-scaling

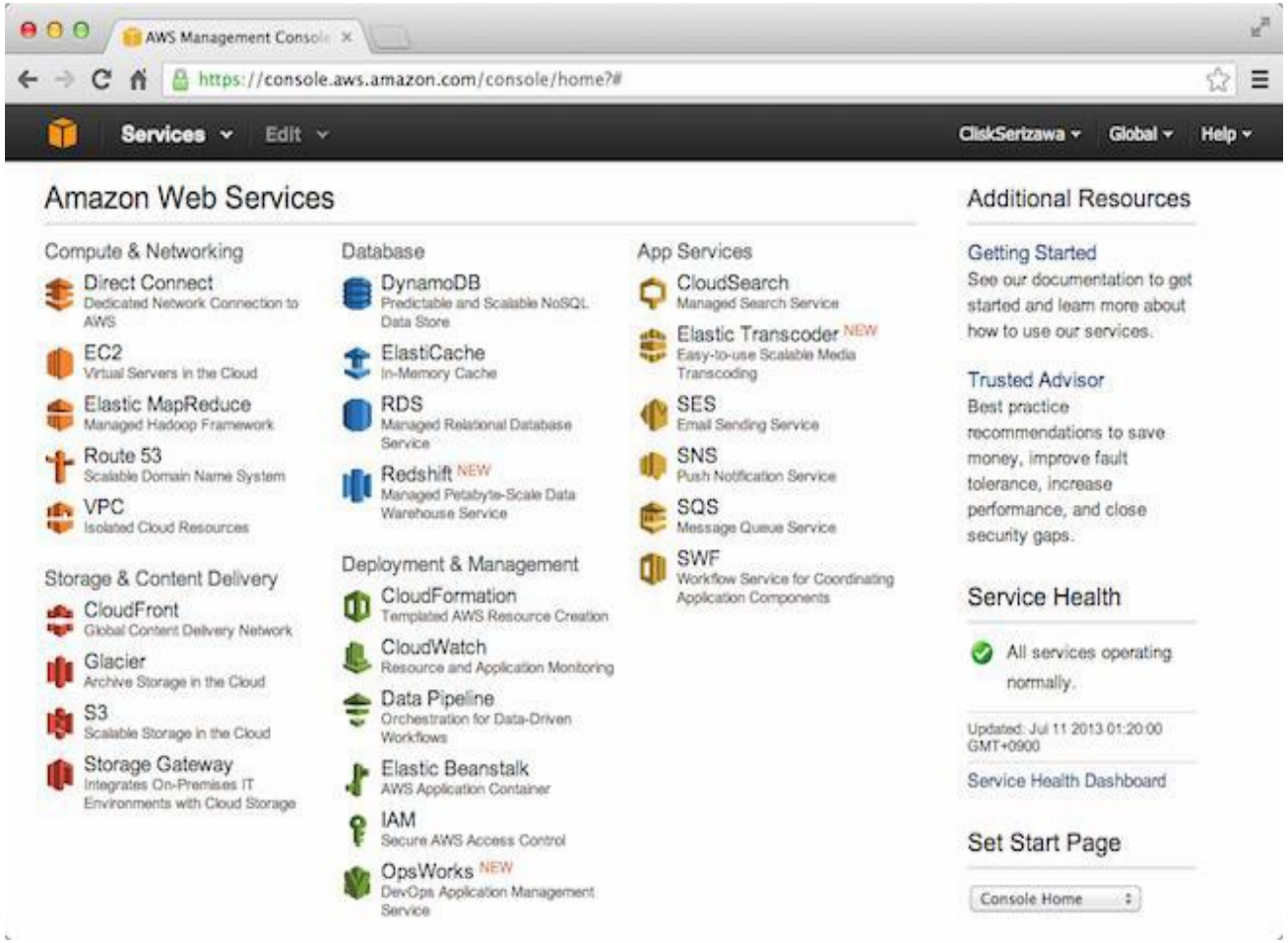


# Self-Service and On-Demand Services

- A consumer can unilaterally provision computing capabilities, such as server time and network storage, as needed automatically without requiring human interaction with each service provider.

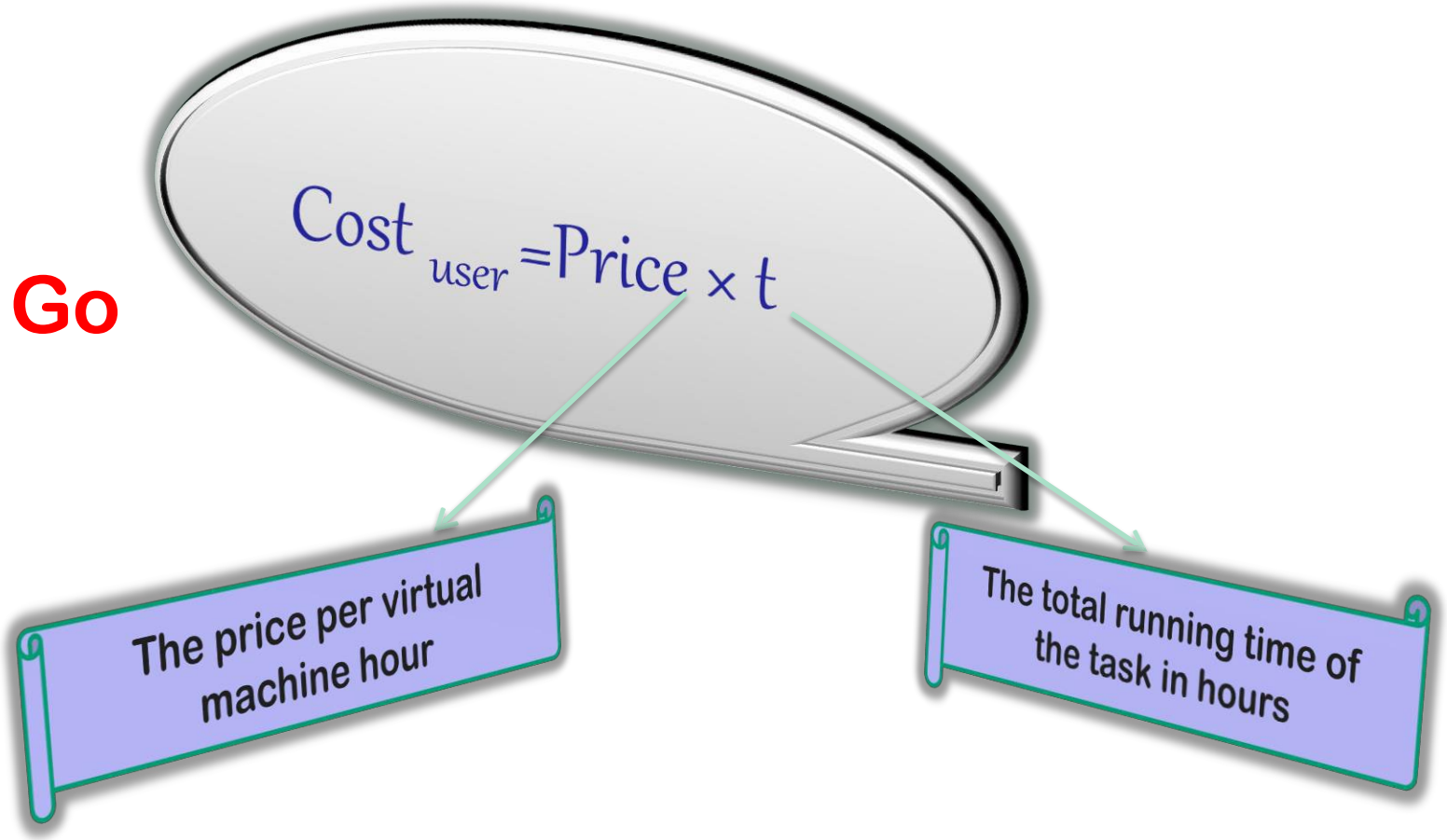
- Example: AWS Management Console



# Pay as you go

## We calculate users' expenses when they execute a task

**Pay as you Go**



# Measured Service

- Cloud systems automatically control and optimize resources by leveraging a metering capability
- Resource usage can be monitored, controlled, and reported, providing transparency for both the provider and consumer of the utilized service.

- Example: Pay-as-you-Go model, auto-scaling

# Cloud service models: IaaS, PaaS, SaaS

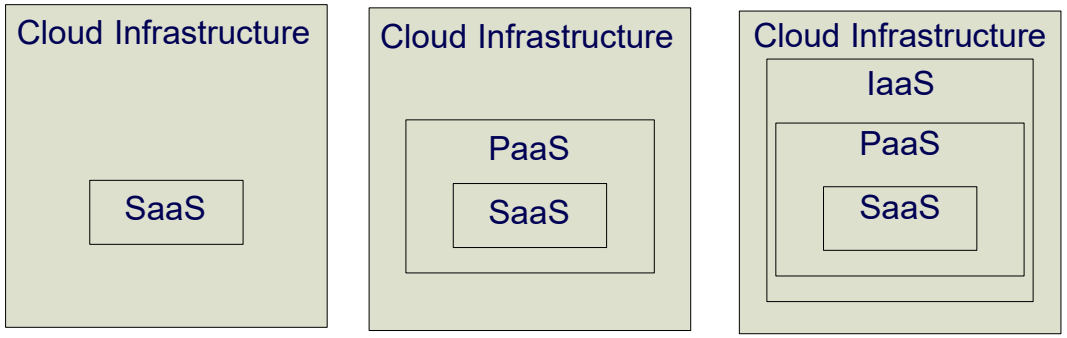
- Cloud Infrastructure as a Service (IaaS)
  - Rent processing, storage, network capacity, and other fundamental computing resources
- Cloud Platform as a Service (PaaS)
  - Deploy customer-created applications to a cloud
- Cloud Software as a Service (SaaS)
  - Use provider's applications over a network
- To be considered "cloud" they must be deployed on top of cloud infrastructure that has the key characteristics

- IaaS
  - Why buy machines when you can rent cycles?
  - Examples: Amazon's EC2, GoGrid, AppNexus
- PaaS
  - Give me nice API and take care of the implementation
  - Example: Google App Engine
- SaaS
  - Just run it for me!
  - Example: Gmail, Facebook

# Cloud Service Model Hierarchy

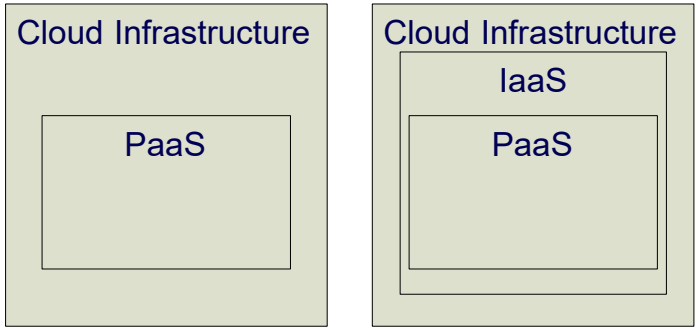
SalesForce CRM

LotusLive

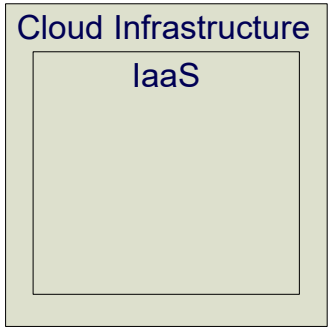



Software as a Service (SaaS) Architectures

Google App




Platform as a Service (PaaS) Architectures




Infrastructure as a Service (IaaS) Architectures

# Cloud Deployment Models

- Private cloud
  - enterprise owned or leased. Cloud infrastructure for single org only, may be managed by the org or a 3rd party, on or off premises
- Public cloud
  - Sold to the public, mega-scale infrastructure. Cloud infrastructure is available to the general public, owned by org selling cloud services
- Hybrid cloud
  - composition of two or more cloud deployment models

# Get your hands dirty with AWS!

- You have now been invited to set up an AWS account.
  - Contact your Helpdesk session tutors if you enrolled late
- Accept and go through Leaner Lab materials to get started

- Enjoy!